

```
\documentclass[11pt]{cls/labreport}

\usepackage{epstopdf}
\usepackage{multicol}
\usepackage[backend=biber,style=apa]{biblatex}
\addbibresource{bib/grn.bib}

\setlength{\columnsep}{1.4em}
\articletype{grn}

\runningtitle{5-node GRN discovery}
\runningauthor{Hector Edu Nseng}

\title{\vspace*{2cm}Discovery of five-node gene regulatory networks for Turing-like pattern formation using PDE/ODE simulation, neural cellular automata, adaptive resonance theory, and symbolic DSL map}

\author[1]{Hector Edu Nseng}
\affil[1]{MRes Systems and Synthetic Biology}

\correspondingauthoraffiliation[{\ast$}]{Correspondence Address: To be completed.}

\keywords{Gene regulatory networks; Turing patterns; neural cellular automata; adaptive resonance theory; symbolic DSL}

\dates{\rec{xx xx, xxxx} \acc{xx xx, xxxx}}

\begin{document}

\inputfile{abstract}
\inputfile{introduction}
\inputfile{background}
\inputfile{methods}
\inputfile{simulator}
\inputfile{nca_art_pipeline}
\inputfile{parameter_search}
\inputfile{results}
\inputfile{discussion}
\inputfile{conclusion}
\inputfile{acknowledgments}
\inputfile{data_availability}
\inputfile{definitions}

\maketitle
\thispagestyle{firststyle}

\begin{multicols}{2}

\begin{abstract}
\abstract
\end{abstract}

\section{Introduction}
\introduction

\section{Background}
```

```
\background

\section{Materials and methods}
\methods

\section{PDE/ODE simulator}
\simulator

\section{NCA-ART discovery pipeline}
\ncaartpipeline

\section{Parameter search and verification}
\parametersearch

\section{Results}
\results

\section{Discussion}
\discussion

\section{Conclusion}
\conclusion

\section{Acknowledgments}
\acknowledgments

\section{Data availability}
\dataavailability

\printbibliography

\end{multicols}
\end{document}
```